SUPERSTORE INTELLIGENCE DASHBOARD

An E-Commerce business intelligence dashboard that turns 4 years of order data into decision senior management can act on this quarter

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01 INTRODUCTION

We built a dashboard to help businesses understand where they make money and where they lose it. Most companies have years of sales data but no easy way to read it. Using 4 years of data from a US superstore (9,983 orders), our dashboard turns those numbers into clear findings that anyone can read and act on.

THE MARGIN GAP WE FOUND

2.34% Furniture Margin vs **17.4%** Technology Margin

Furniture sells almost as much as Technology ($736,880 vs $834,554) but keeps a fraction of the profit that exposing a pricing problem most spreadsheets never surface

Total Sales	$2.29M
Total Profit	$284K
Order Analyzed	9, 983
Profit Margin	12.4%

02 THE PROBLEM

Sales, profits, and discount data sit in disconnected spreadsheets. Managers cannot see which product lines lose money without manually rebuilding pivot tables every time.

03 WHO IT SERVES

- Senior Management - Strategic Overview
- Regional Managers - Category Drill - Down
- Finance Team - Discount Policy Impact
- General Staff - Plain - Language Read

04 BUILT WITH

Streamlit (Python) + Plotly, chosen so the whole team works in one language and integrates directly with pandas for cleaning and aggregation.

05 USERS

EARLY CHILDHOOD - FIRST TIME USER

Simple View

- Big icon button replace filter menus
- Only 3 plain-language KPI cards
- Advanced charts hidden to avoid clutter
- 18px base font, large tap targets

ADULT - ANALYST USER

Standard View

- 5 filters that cross-combine instantly
- Full 5-card KPI row include margin & discount
- Drill-down, heat map, parallel coordinates
- Auto-generated written insight panel

ELDERLY - ACCESSIBILITY - FOCUSED

Large Text View

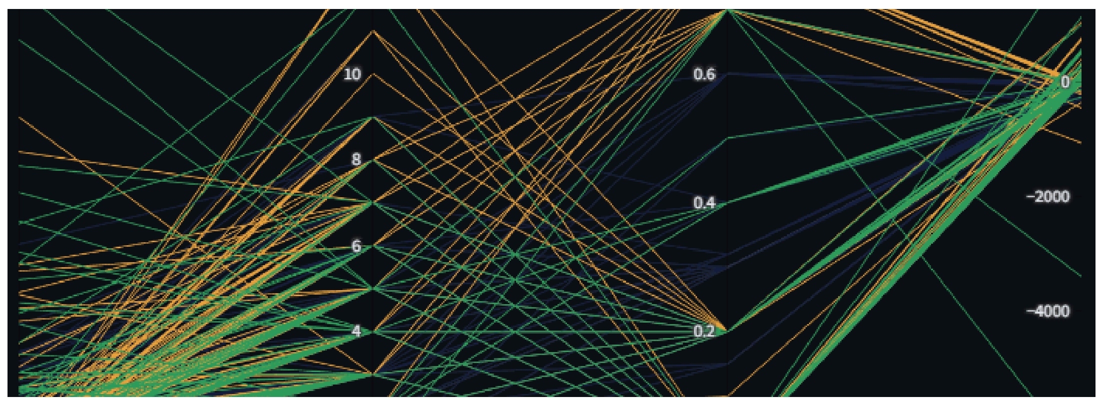
- 21px base font across every label
- Same full chart depth as Standard View
- Identical navigation
- Larger click targets throughout

06 DASHBOARD

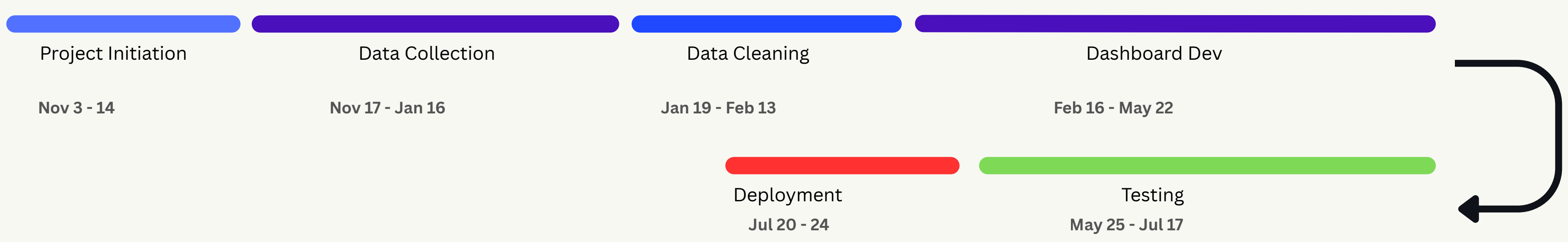
Parallel Coordinates

Sales, Quantity, Discount and Profit shown as one connected relationship across thousands of orders.

Orders with higher discounts, particularly **30%** and above, are more likely to result in negative profit, as shown by the lines crossing into the negative profit region.



Project Timeline (Nov 2025 - Jul 2026)



Profit Margin Heat Map

Region x Category compressed into one scannable grid. A 12 bar comparison no bar chart handles cleanly

	Furniture	Office	Tech
Central	-1.8	5.3	19.8
East	0.9	19.9	17.9
South	5.8	15.9	13.4
West	4.6	23.8	17.6

West holds the strongest margin at **15.3%** while Central is weakest at **7.8%** that invisible in a single combined total.

Key Insight Generated

-$17, 725

Tables is the largest loss - making sub - category, averaging a 26% discount. The likely cause

2 States

Texas & Pennsylvania rank top - 5 in sales yet post a net loss, priority candidates for pricing review

13.5%

Of all orders used a **30%+ discount** and still lost money. That is a clear threshold to revisit

07 CONCLUSION

Three things stood out from the data. Furniture barely makes a profit at 2.34% margin. Tables loses money because of high discounts. Texas and Pennsylvania sell a lot but still lose money overall. Our dashboard finds these problems automatically and updates every time you change a filter. Anyone can use it, from a first-time user to a senior manager.